

3.6 Alternating Series Worksheet

Determine whether each of the following series converge or diverge. You may use any technique from 3.6 or earlier.

$$1. \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln(n+4)}$$

We see that this is an alternating series with $a_n = \frac{1}{\ln(n+4)}$. Clearly a_n is decreasing and $\lim_{n \rightarrow \infty} a_n = 0$.

Therefore, by the AST, this series converges. ■

$$2. \sum_{k=1}^{\infty} \frac{(-3)^{k+1}}{2^k}$$

We first rewrite the series as $\sum_{k=1}^{\infty} \frac{(-3)^{k+1}}{2^k} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} 3^{k+1}}{2^k}$. Therefore, this is an alternating series, but we should also note that this is a geometric series and so we further rewrite the series as

$$\sum_{k=1}^{\infty} \frac{(-3)^{k+1}}{2^k} = \sum_{k=1}^{\infty} \frac{-3(-3)^k}{2^k} = \sum_{k=1}^{\infty} -3 \left(\frac{-3}{2} \right)^k = \sum_{k=1}^{\infty} \frac{9}{2} \left(\frac{-3}{2} \right)^{k-1}$$

We see that $r = -\frac{3}{2}$ and since $|r| \geq 1$ this series diverges. ■

$$3. \sum_{n=1}^{\infty} \frac{2n}{n^2 + 1}$$

We see that $\frac{2n}{n^2 + 1} > \frac{2n}{n^2} = \frac{2}{n} > \frac{1}{n}$. Clearly the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is the Harmonic series and diverges, so we can't reach a conclusion here via the Comparison Test. However, we can proceed with using the LCT

test. Note that both series contain only positive terms. Note that $\lim_{n \rightarrow \infty} \frac{\left(\frac{2n}{n^2 + 1} \right)}{\left(\frac{1}{n} \right)} = \lim_{n \rightarrow \infty} \frac{2n^2}{n^2 + 1} = 2$. Since

the result is both positive and finite we know that both series have the same behavior. Therefore, this series must diverge. ■

Note: This question also could have been answered using the Integral Test, or the LCT or a variety of other tests.

4. $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2 + 2}$

We note that this is an alternating series with $a_n = \frac{n}{n^2 + 2}$. We note that

$$\lim_{n \rightarrow \infty} \frac{n}{n^2 + 2} = \lim_{n \rightarrow \infty} \frac{n}{n^2 + 2} \cdot \frac{1/n}{1/n} = \lim_{n \rightarrow \infty} \frac{1}{n + \frac{2}{n}} = 0.$$

$$\text{Also } \frac{a_{n+1}}{a_n} = \frac{\left(\frac{n+1}{(n+1)^2 + 2} \right)}{\left(\frac{n}{n^2 + 2} \right)} = \frac{n+1}{n^2 + 2n + 3} \cdot \frac{n^2 + 2}{n} = \frac{n^3 + n^2 + 2n + 2}{n^3 + 2n^2 + 3} = \frac{(n^3 + n^2 + 2) + n}{(n^3 + n^2 + 2) + n^2 + 1} < 1$$

So clearly a_n is decreasing.

Therefore, by the AST, this series converges. ■

5. $\sum_{n=1}^{\infty} (-1)^n e^{2/n}$

This is clearly an alternating series with $a_n = e^{2/n}$. We see that $\frac{2}{n}$ is clearly decreasing and because the function $y = e^x$ is always increasing this means that a_n is decreasing. We also note that $\lim_{n \rightarrow \infty} e^{2/n} = e^0 = 1$.

Because we didn't get a limit result of 0 here the AST is not helpful. However, we do then note the behavior of the even and odd terms of the series.

Odd terms: $a_n = -e^{2/n}$ and so $\lim_{n \rightarrow \infty} a_n = -1$.

Even terms: $a_n = e^{2/n}$ and so $\lim_{n \rightarrow \infty} a_n = 1$.

Since the even and odd terms don't approach the same value we can say that $\lim_{n \rightarrow \infty} a_n$ DNE and therefore, by the Test for Divergence, this series diverges. ■

6. $-\frac{1}{1} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} - \frac{1}{5^2} + \frac{1}{6^2} - \dots$ where there are always two negative terms followed by a positive term.

Note that this series doesn't have an easy series formula, but we note that if we were to absolute value all terms of the series then we would have $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{1}{n^2}$ and this is a p-series with $p = 2 > 1$ and therefore converges. This means that the given series is absolutely convergent and therefore is it convergent. ■

7. The series $\sum_{n=1}^{\infty} (e^{-n^2} - e^{-(n+1)^2})$ converges. Determine what value the series converges to.

We suspect this series is telescoping and so we attempt to find s_k .

$$s_k = (e^{-1} - e^{-4}) + (e^{-4} - e^{-9}) + (e^{-9} - e^{-16}) + \dots + (e^{-k^2} - e^{-(k+1)^2}) = e^{-1} - e^{-(k+1)^2}.$$

Note that $\lim_{k \rightarrow \infty} (e^{-1} - e^{-(k+1)^2}) = \lim_{k \rightarrow \infty} \left(\frac{1}{e} - \frac{1}{e^{(k+1)^2}} \right) = \frac{1}{e} - 0 = \frac{1}{e}.$

Therefore $\sum_{n=1}^{\infty} (e^{-n^2} - e^{-(n+1)^2}) = \frac{1}{e}$

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8. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2n+1}{3n-1}$

This is clearly an alternating series with $a_n = \frac{2n+1}{3n-1}$. We note that $\lim_{n \rightarrow \infty} \frac{2n+1}{3n-1} = \frac{2}{3}$. Because we didn't get a limit result of 0 here the AST is not helpful. However, we do then note the behavior of the even and odd terms of the series.

Odd terms: $a_n = \frac{2n+1}{3n-1}$ and so $\lim_{n \rightarrow \infty} a_n = \frac{2}{3}$

Even terms: $a_n = -\frac{2n+1}{3n-1}$ and so $\lim_{n \rightarrow \infty} a_n = -\frac{2}{3}$

Since the even and odd terms don't approach the same value we can say that $\lim_{n \rightarrow \infty} a_n$ DNE and therefore, by the Test for Divergence, this series diverges.

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Classify each of the following series as absolutely convergent, conditionally convergent, or divergent.

9. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n+1}$

We first check to see if the given series is absolutely convergent. We note that $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{3n+1} \right| = \sum_{n=1}^{\infty} \frac{1}{3n+1}$.

Clearly $\frac{1}{3n+1} \geq \frac{1}{3n+n} = \frac{1}{4n}$ for all $n \geq 1$ and $\sum_{n=1}^{\infty} \frac{1}{4n} = \frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n}$ which is the harmonic series and therefore diverges. So by the Comparison test the series $\sum_{n=1}^{\infty} \frac{1}{3n+1}$ must also diverge. So the series is not absolutely convergent.

We will now use the alternating series test to see if the given series converges conditionally.

We let $a_n = \frac{1}{3n+1}$. We see that $a_{n+1} < a_n$ for all n . Also, $\lim_{n \rightarrow \infty} a_n = 0$. Therefore by the alternating series

test the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n+1}$ converges.

So $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n+1}$ is conditionally convergent.

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10. $\sum_{k=1}^{\infty} \frac{\cos(k)}{k^2}$

We first check to see if the given series is absolutely convergent. We note that

$\sum_{k=1}^{\infty} \left| \frac{\cos(k)}{k^2} \right| = \sum_{k=1}^{\infty} \frac{|\cos(k)|}{k^2} \leq \sum_{k=1}^{\infty} \frac{1}{k^2}$ which is a p -series with $p = 2 > 1$ and it therefore converges. Therefore,

by the comparison theorem the series $\sum_{k=1}^{\infty} \left| \frac{\cos(k)}{k^2} \right|$ is convergent. So, $\sum_{k=1}^{\infty} \frac{\cos(k)}{k^2}$ is absolutely convergent.

■

$$11. \sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{2k^3 + 1}$$

We first check to see if the given series is absolutely convergent. We note that $\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1} k}{2k^3 + 1} \right| = \sum_{k=1}^{\infty} \frac{k}{2k^3 + 1}$.

Consider the series $\sum_{k=1}^{\infty} \frac{1}{k^2}$ which is a convergent p -series with $p = 2 > 1$. We note that both series contain

only positive terms. We calculate: $\lim_{k \rightarrow \infty} \frac{\left(\frac{k}{2k^3 + 1} \right)}{\left(\frac{1}{k^2} \right)} = \lim_{k \rightarrow \infty} \frac{k^3}{2k^3 + 1} = \frac{1}{2}$.

Since the limit is a positive, finite number we know, by the Limit Comparison test, that $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{2k^3 + 1}$

converges. So $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{2k^3 + 1}$ is absolutely convergent. ■

$$12. \sum_{n=1}^{\infty} \frac{(-1)^n (n+3)}{n}$$

We note that the odd terms of the series are negative and the even terms of the series are positive. We also see that $\lim_{n \rightarrow \infty} \frac{(n+3)}{n} = 1$ and therefore the odd terms of the series converge towards -1 and the even terms of

the series converge towards 1. Therefore $\lim_{n \rightarrow \infty} \frac{(-1)^n (n+3)}{n}$ DNE and so by the Test for Divergence the

series $\sum_{n=1}^{\infty} \frac{(-1)^n (n+3)}{n}$ must diverge. ■

$$13. \sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n+3}}{n}$$

We first check to see if the given series is absolutely convergent. We note that

$\sum_{n=1}^{\infty} \left| \frac{(-1)^n \sqrt{n+3}}{n} \right| = \sum_{n=1}^{\infty} \frac{\sqrt{n+3}}{n}$. Consider the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ which is a divergent p -series with $p = \frac{1}{2} < 1$. We

note that both series contain only positive terms. We calculate:

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{\sqrt{n+3}}{n} \right)}{\left(\frac{1}{\sqrt{n}} \right)} = \lim_{n \rightarrow \infty} \frac{\sqrt{n^2+3n}}{\sqrt{n^2}} = \lim_{n \rightarrow \infty} \sqrt{\frac{n^2+3n}{n^2}} = \lim_{n \rightarrow \infty} \sqrt{1 + \frac{3}{n}} = 1.$$

Since the limit is a positive, finite number we know, by the Limit Comparison test, that

$\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n+3}}{n}$ diverges. So

$\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n+3}}{n}$ is not absolutely convergent.

We will now use the alternating series test to see if the given series converges conditionally.

We let $b_n = \frac{\sqrt{n+3}}{n}$. We see that $\frac{b_{n+1}}{b_n} = \frac{\frac{\sqrt{n+4}}{n+1}}{\frac{\sqrt{n+3}}{n}} = \frac{n}{n+1} \sqrt{\frac{n+4}{n+3}} = \sqrt{\frac{n^2(n+4)}{(n+1)^2(n+3)}} = \sqrt{\frac{n^3+4n^2}{n^3+5n^2+7n+3}} < 1$

for all $n \geq 1$. Therefore we know that b_n is decreasing. Also,

$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{\sqrt{n+3}}{n} = \lim_{n \rightarrow \infty} \frac{\sqrt{n+3}}{\sqrt{n^2}} = \lim_{n \rightarrow \infty} \sqrt{\frac{n+3}{n^2}} = \lim_{n \rightarrow \infty} \sqrt{\frac{1}{n} + \frac{3}{n^2}} = 0$. Therefore by the alternating series test the

series $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n+3}}{n}$ converges.

So $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n+3}}{n}$ is conditionally convergent.

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14. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^6}$ converges by the Alternating Series Test.

- a) Determine the upper bound on how far off the 5th partial sum of the series is from the actual sum of the series.

We note that for this series $a_n = \frac{1}{n^6}$. If s is the sum of the series we know that $|s - s_5| \leq a_6 = \frac{1}{6^6}$.

Therefore, we know that the 5th partial sum of the series within $\frac{1}{6^6} \approx 0.00002$ of the actual sum.

- b) How many terms of the series are needed in order to find the sum of the series to within 0.00005.

We know that

$|s - s_n| \leq a_{n+1} = \frac{1}{(n+1)^6}$. We want to find a value of n that makes $\frac{1}{(n+1)^6} < 0.00005$.

We see that $\frac{1}{(n+1)^6} < 0.00005 \Rightarrow 20000 < (n+1)^6 \Rightarrow \sqrt[6]{20000} < n+1 \Rightarrow n > \sqrt[6]{20000} - 1 \approx 4.21$

So the 5th partial sum is the first partial sum that we can guarantee is within one 0.00005 of the actual sum s .

15. The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n \cdot 5^n}$ converges by the Alternating Series Test.

- a) Determine the upper bound on how far off the 2nd partial sum of the series is from the actual sum of the series.

We note that for this series $a_n = \frac{1}{n5^n}$. If s is the sum of the series we know that

$|s - s_2| \leq a_3 = \frac{1}{3 \cdot 5^3} = \frac{1}{375}$. Therefore, we know that the 2nd partial sum of the series within $\frac{1}{375} \approx 0.003$ of the actual sum.

- b) How many terms of the series are needed in order to find the sum of the series to within 0.0001.

We know that

$|s - s_n| \leq a_{n+1} = \frac{1}{(n+1)5^{n+1}}$. We want to find a value of n that makes $\frac{1}{(n+1)5^{n+1}} < 0.00005$. We will do

this by testing values rather than solving for n since this inequality is quite difficult.

We see that $\frac{1}{(n+1)5^{n+1}} < 0.0001 \Rightarrow 10000 < (n+1)5^{n+1} \Rightarrow n \geq 5$

So the 5th partial sum is the first partial sum that we can guarantee is within 0.0001 of the actual sum s .

16. The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^4}$ converges by the Alternating Series Test.

- a) Determine an upper bound on how far off the 10th partial sum of the series is from the actual sum of the series.

If s is the sum of the series we know that $|s - s_{10}| \leq a_{11} = \frac{1}{11^4}$. Therefore we know that the 10th partial sum of the series is within $\frac{1}{11^4} \approx 0.0001$ of the actual sum.

- b) What is the first partial sum of this series that we can guarantee is within one thousandth of the actual sum?

Since we know that $|s - s_n| \leq a_{n+1} = \frac{1}{(n+1)^4}$ we just want to find what the first value of n is that

makes $\frac{1}{(n+1)^4} < \frac{1}{1000}$.

Clearly, $\frac{1}{(n+1)^4} < \frac{1}{1000} \Rightarrow 1000 < (n+1)^4 \Rightarrow n+1 > \sqrt[4]{1000} \Rightarrow n = \sqrt[4]{1000} - 1 \approx 4.6234$

So the 5th partial sum is the first partial sum that we can guarantee is within one thousandth of the actual sum s .

17. The series $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n+3}}{n}$ converges by the Alternating series test.

- a) If this sum is estimated by the 21st partial sum, determine an upper bound for the error on this approximation.

If s is the sum of the series we know that $|s - s_{21}| \leq a_{22} = \frac{\sqrt{22+3}}{22} = \frac{5}{22}$. Therefore we know that the 21st partial sum of the series is within $\frac{5}{22} \approx 0.2273$ of the actual sum.

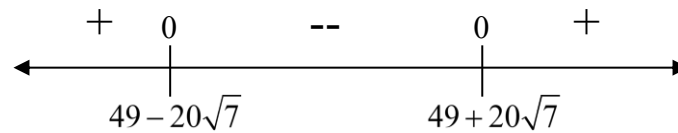
- b) What is the first partial sum of this series that we can guarantee is within one tenth of the actual sum?

Since we know that $|s - s_n| \leq a_{n+1} = \frac{\sqrt{n+4}}{n+1}$ we just want to find what the first value of n is that

makes $\frac{\sqrt{n+4}}{n+1} < \frac{1}{10}$.

Clearly, $\frac{\sqrt{n+4}}{n+1} < \frac{1}{10} \Rightarrow 10\sqrt{n+4} < n+1 \Rightarrow 100(n+4) < (n+1)^2 \Rightarrow n^2 - 98n - 399 > 0$

By the Quadratic formula $n^2 - 98n - 399 = 0$ when $n = 49 \pm 20\sqrt{7}$. We then test for the behavior of the quadratic.



Since we are interested in when $n^2 - 98n - 399 > 0$ we can see that this occurs for $n > 49 + 20\sqrt{7} \approx 101.9150$. So the 102nd partial sum is the first partial sum that we can guarantee is within one tenth of the actual sum s .